# Critical Collapse of a Massless Scalar Field with Brill Wave Perturbations in Axisymmetry

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In the parameter space near the threshold of black hole formation numerical analysis reveals a number of "critical phenomena" in the resulting spacetimes. These solutions have proven to be a powerful test of cosmic censorship, but are less understood in higher-dimensional systems which pose computational challenges. Using the ADM decomposition, Z4 formulation, and Geroch decomposition, we build an wavelet-adaptive, finite-difference code for simulating axisymmetric critical collapse. To probe the impact of gravitational waves on higher dimensional critical phenomena, we consider the case of a spherically symmetric, massless scalar field superimposed with a weak axisymmetric Brill wave. We find that gravitational wave perturbations decay as scalar field dynamics become dominant near the critical point and that the gravitational wave contributes constructively to scalar field critical collapse.

## I. INTRODUCTION

One of the pivotal results of Numerical Relativity was the discovery of intriguing structure in certain spacetimes near the threshold of black hole formation. Choptuik's pioneering paper in this field observed several kinds of "critical phenomena" in the gravitational collapse of spherically symmetric massless scalar fields of varying amplitude: power law mass scaling, universality, and self-similarity [1]. These phenomena occur due to the *strong-field* dynamics of critical evolution, and are therefore prominent in regions where strong-field dynamics dominate, and reflect the non-linear character of General Relativity (GR). Notably, the mass scaling relation implies there exists a measure-zero point in the parameter space in which the solution gives rise to a naked singularity [2].

Extending these studies to less symmetric systems poses substantial computational challenges, as resolving the fine-scale structure near the black hole threshold requires extremely high spatial and temporal resolution. Of particular interest is the case of vacuum collapse: the formation of black holes from pure gravitational waves (Brill waves [3] or Teukolsky waves [4] are commonly used to model vacuum collapse). All asymmetric systems generate gravitational waves and their effect on critical phenomena is not completely understood, as studies have yielded conflicting results when it comes to vacuum collapse. Abraham and Evans reported mass-law scaling, universality, and self-similarity in axisymmetric Teukolsky wave collapse [5]. However, more recent studies cast doubt on these results and especially on the existence of universality in vacuum collapse [6, 7].

One available avenue of exploration is superimposing multiple energy-momentum sources on top of one another to better understand how gravitational wave perturbations evolve in higher dimensions. One such study used spherically symmetric matter fields to simulate the two wave-like degrees of freedom present in axisymmetric gravitational waves, and found that scalar field dynamics dominate the system's evolution when sufficiently fine-tuned [8]. A recent study found that, in a highly symmetric 4+1 dimensional extension of GR, weak gravitational wave perturbations to a massless scalar field decay over time [9]. Finally, the authors of [10, 11] found that during near-critical massless and massive scalar field evolution, inter-field interaction has an inhibiting (rather than constructive) effect on criticality [10], and that each energy-momentum source could be dominant near a critical "triple-point" in parameter space.

In this article we consider the case of scalar field evolution with a weak Brill wave perturbation in axisymmetry to determine which dynamics dominate near criticality. This serves as both a direct exploration of how the introduction of gravitational waves affects scalar field evolution in an axisymmetric decomposition of GR, as well as the continuation of previous competing collapse studies [10, 11]. To this end, this work provides an initial exploration of the available parameter space and was built with future competing collapse scenarios in mind.

## II. MATHEMATICAL FORMALISM

Suppose $\mathcal{M}$ is a four-dimensional Lorentzian manifold with metric $g_{ab}$, covariant derivative operator $\nabla_a$, and a preferred cylindrical coordinate chart $(t, r, z, \phi)$. To reduce the Einstein field equations into a form suitable for axisymmetric evolution we use a combination of mathematical techniques.

### A. ADM Decomposition

First we perform the standard ADM decomposition [12] using similar notation to Baumgarte and Shapiro

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[13]. This decomposition partitions $\mathcal{M}$ into a set of non-intersecting spacelike 3-surfaces $\Sigma(t)$, which may be interpreted as the level surfaces of some scalar function $t$ with unit normal $n^a = -\alpha\nabla^a t$, where $\alpha$ is a normalization factor. We assume that these 3-surfaces are themselves submanifolds of $\mathcal{M}$ and can thus define *spatial* tensors which reside solely in $\Sigma(t)$. This includes the spatial metric $\gamma_{ab}$ and the extrinsic curvature $K_{ab}$.

$$\gamma_{ab} \equiv g_{ab} + n_a n_b, \tag{1}$$

$$K_{ab} \equiv -\gamma_a{}^c \gamma_b{}^d \nabla_c n_d, \tag{2}$$

which in turn induces a spatial covariant derivative operator and a spatial Riemann tensor

$$\hat{\nabla}_a T_b = \gamma_a{}^c \gamma_b{}^d \nabla_c (\gamma_d{}^e T_e), \tag{3}$$

$$\hat{R}^d{}_{cab} V_d = 2\hat{\nabla}_{[a}\hat{\nabla}_{b]} V_c. \tag{4}$$

We also decompose $T_{ab}$ along $n^a$.

$$\begin{aligned} \rho &\equiv n^a n^b T_{ab} \\ J_a &\equiv -\gamma_a{}^c n^d T_{cd} \\ S_{ab} &\equiv \gamma_a{}^c \gamma_b{}^d T_{cd}. \end{aligned} \tag{5}$$

To introduce evolution between spatial surfaces we define the temporal four-vector

$$t^a \equiv \alpha n^a + \beta^a \tag{6}$$

where $\beta^a$ is an arbitrary spatial vector. Here $t^a$ can be thought of as connecting points with the same spatial coordinates on neighboring time slices and $\alpha$ and $\beta^a$ are gauge freedoms (the lapse function and shift vector respectively). Thus quantities evolve as $\partial_t \equiv \alpha\mathcal{L}_{\mathbf{n}} + \mathcal{L}_\beta$. By fixing our zeroth coordinate vector to lie along $t^\alpha$ and the other three coordinate vectors to be spatial, we can draw a one-to-one correspondence between spatial *four*-tensors on $\mathcal{M}$ and *three*-tensors on $\Sigma(t)$. This allows us to compute derivatives and geometric quantities like $\hat{\Gamma}^c{}_{ab}$ and $\hat{R}_{ab}$ directly in $\Sigma(t)$, thus justifying our switch to 3-dimensional indices in the following sections.

Altogether the ADM decomposition yields two constraint equations

$$\hat{R} + K^2 - K^i{}_j K^j{}_i = 2\kappa\rho, \tag{7}$$

$$\hat{\nabla}_j K^j{}_i - \hat{\nabla}_i K = \kappa J_i, \tag{8}$$

and two evolution equations

$$\mathcal{L}_{\mathbf{n}}\gamma_{ij} = -2K_{ij}, \tag{9}$$

$$\begin{aligned} \mathcal{L}_{\mathbf{n}} K^i{}_j =& -\alpha^{-1}\hat{\nabla}^i\hat{\nabla}_j\alpha + \hat{R}^i{}_j + KK^i{}_j \\ & - \kappa\left(S^i{}_j - \frac{1}{2}\gamma^i{}_j(S-\rho)\right). \end{aligned} \tag{10}$$

### B. Z4 Framework

By default, the system of equations resulting from the ADM formulations suffers from instability in higher dimensions. To avoid this, we utilize the Z4 framework, wherein a fictitious 4-vector $Z^a$ encoding numerical error is introduced to the field equations, ultimately allowing us to transform the elliptic constraint equations to a fully hyperbolic system of PDEs. The field equations become

$$R_{ab} + 2\nabla_{(a}Z_{b)} = \kappa\left(T_{ab} - \frac{1}{2}T g_{ab}\right), \tag{11}$$

with projections

$$\begin{aligned} \Theta &= -n_a \tilde{Z}^a \\ \hat{Z}^a &= \gamma^a{}_c Z^c. \end{aligned} \tag{12}$$

We can absorb this four vector into a modified energy-momentum component as

$$\tilde{T}_{ab} = T_{ab} - 2/\kappa\left(\nabla_{(a}Z_{b)} - \frac{1}{2}g_{ab}\nabla_c Z^c\right) \tag{13}$$

so

$$\tilde{\rho} = \rho + \frac{1}{\kappa}\left(\mathcal{L}_n\Theta + \hat{Z}_b a^b - \hat{\nabla}_a\hat{Z}^a + K\Theta\right). \tag{14}$$

$$\hat{J}_a = J_a + \frac{1}{\kappa}\left(\mathcal{L}_n\hat{Z}_a + 2K_{ab}\hat{Z}^b + \Theta a_a - \hat{\nabla}_a\Theta\right) \tag{15}$$

$$\tilde{S}_{ab} = S_{ab} - \frac{2}{\kappa}\gamma_a{}^c\gamma_b{}^d\left(\nabla_{(c}Z_{d)} - \frac{1}{2}g_{cd}\nabla_e Z^e\right) \tag{16}$$

$$\tilde{S} = S + \frac{1}{\kappa}\left(\hat{\nabla}_c\hat{Z}^c - K\Theta + 3\hat{Z}_c a^c + 3\mathcal{L}_n\Theta\right). \tag{17}$$

We can insert these modified terms into Eqs. (9)-(10) to find evolution equations for $\gamma_{ij}$ and $K_{ij}$ and into Eqs. (7)-(8) to derive constraint evolution

$$\begin{aligned} \mathcal{L}_{\mathbf{n}}\Theta =& \frac{1}{2}\left(\hat{R} + K^2 - K^{ij}K_{ij}\right) - \kappa\rho \\ & - K\Theta + \hat{\nabla}_i\hat{Z}^i - \hat{Z}_i a^i \\ \mathcal{L}_{\mathbf{n}}\hat{Z}_i =& \hat{\nabla}^j\left(K_{ij} - \gamma_{ij}K\right) - \kappa J_i \\ & + \hat{\nabla}_i\Theta - \Theta a_i - 2K_{ij}\hat{Z}^j \end{aligned} \tag{18}$$

### C. Geroch Decomposition

Finally, in order to reduce our system of equations to the axisymmetric case, we introduce a static spatial Killing vector with closed orbits $\xi^i \equiv \left(\frac{\partial}{\partial\phi}\right)^i$. Letting $\mathcal{N}(t)$ be the collection of orbits of $\xi^i$, there is a one-to-one correspondence between tensors on $\mathcal{N}(t)$ and tensors

in $\Sigma(t)$ that are orthogonal to $\xi^i$ and have vanishing Lie derivative along $\xi^i$. We call such tensors *axisymmetric*.

Decomposing spacetime along this Killing vector is known as the Geroch decomposition [14]. We assume all relevant tensors (i.e. $\gamma_{ij}, K_{ij}, \Theta, \hat{Z}_i, \alpha, \beta_i, \rho, J_i, S_{ij}$) have vanishing Lie derivative along $\xi^i$ and introduce the norm of the Killing vector

$$\lambda^2 = \gamma_{ij}\xi^i\xi^j > 0 \tag{19}$$

the induced axisymmetric metric

$$h_{ij} = \gamma_{ij} - \lambda^{-2}\xi_i\xi_j \tag{20}$$

the induced axisymmetric derivative

$$D_i T^j{}_k = h_i{}^l \hat{\nabla}_l (h^j{}_m h_k{}^n T^m{}_n). \tag{21}$$

and the natural projection of the Riemann curvature tensor

$$\mathcal{R}^l{}_{kij} V_l = 2D_{[i} D_{j]} V_k. \tag{22}$$

Finally, we assume that our Killing vector is twist-free in $\mathcal{M}$

$$\epsilon_{abcd}\xi^b \nabla^c \xi^d = 0 \tag{23}$$

which implies that the angular component of the Z-vector vanishes $\xi^i Z_i = 0$, as do the angular momentum and angular shear terms of the energy momentum tensor.

$$\begin{aligned} \xi^i J_i &= 0 \\ h_i{}^j \xi^k S_{jk} &= 0 \end{aligned} \tag{24}$$

Using Rinne's derivation [15], twist-free axisymmetry allows us to relate the two-dimensional Ricci curvature tensor to its three-dimensional counterpart via

$$\begin{aligned} h_i{}^k h_j{}^l \hat{R}_{kl} &= \mathcal{R}_{ij} - \frac{D_i D_j \lambda}{\lambda} \\ \lambda^{-1} h_i{}^k \xi^l \hat{R}_{kl} &= 0 \\ \lambda^{-2} \xi^i \xi^j \hat{R}_{ij} &= -\frac{D_i D^i \lambda}{\lambda} \end{aligned} \tag{25}$$

Similarly to the ADM decomposition, we can now decompose various spatial tensors into their purely axisymmetric counterparts. The non-zero projections of the extrinsic curvature are

$$\begin{aligned} X_{ij} &= h_i{}^k h_j{}^l K_{kl} \\ L &= \lambda^{-2} \xi^i \xi^j K_{ij}. \end{aligned} \tag{26}$$

For $\hat{Z}^i$ this becomes

$$\mathcal{Z}_i = h_i{}^j \hat{Z}_j. \tag{27}$$

Finally, for energy-momentum we have

$$\begin{aligned} \mathcal{J}_i &= h_i{}^k J_k \\ \tau &= \lambda^{-2} \xi^i \xi^j S_{ij} \\ \mathcal{S}_{ij} &= h_i{}^k h_j{}^l S_{kl}. \end{aligned} \tag{28}$$

Rewriting our results from the previous section in terms of these decompositions we arrive at the following equations. Similarly to the ADM decomposition, we can set the third coordinate vector to $\xi^i$ and assume the other two spatial vectors are axisymmetric. This justifies our shift to 2-dimensional indices, which are denoted by uppercase Latin letters. Our final evolution equations are as follows:

$$\begin{aligned} \mathcal{L}_{\mathbf{n}} h_{AB} =& -2X_{AB} \\ \mathcal{L}_{\mathbf{n}} \lambda =& -\lambda L \\ \mathcal{L}_{\mathbf{n}} X_{AB} =& \mathcal{R}_{AB} - \lambda^{-1} D_A D_B \lambda - \alpha^{-1} D_A D_B \alpha \\ &+ (X+L) X_{AB} - 2X_A{}^C X_{CB} \\ &- \kappa \left( \mathcal{S}_{AB} + \frac{1}{2} h_{AB} (\rho - \mathcal{S} - \tau) \right) \\ &+ 2D_{(A} \mathcal{Z}_{B)} - 2X_{AB}\Theta \\ \mathcal{L}_{\mathbf{n}} L =& -\lambda^{-1} D_A D^A \lambda - (\alpha\lambda)^{-1} D_A \lambda D^A \alpha \\ &+ (X+L)L - \frac{\kappa}{2}(\rho - \mathcal{S} + \tau) \\ &+ 2\mathcal{Z}^A \lambda^{-1} D_A \lambda - 2L\Theta \\ \mathcal{L}_{\mathbf{n}} \Theta =& \frac{1}{2}(X^2 - X_{AB}X^{AB} + \mathcal{R}) \\ &- \frac{D_A D^A \lambda}{\lambda} + XL - \kappa\rho \\ &+ (\lambda^{-1} D_A \lambda - \alpha^{-1} D_A \alpha) \mathcal{Z}^A \\ &+ D_A \mathcal{Z}^A - (X+L)\Theta \\ \mathcal{L}_{\mathbf{n}} \mathcal{Z}_A =& D_B X_A{}^B - D_A (X+L) \\ &+ \lambda^{-1} X_{AB} D^B \lambda - \lambda^{-1} L D_A \lambda - \kappa \mathcal{J}_A \\ &+ 2X_{AB} \mathcal{Z}^B - \alpha^{-1} \Theta D_A \alpha + D_A \Theta \end{aligned} \tag{29}$$

### D. Regularity and Boundary Conditions

We must carefully treat the axis of symmetry to avoid instabilities quickly forming at this coordinate singularity. To this end we enforce regularity conditions derived in [15]. Scalar fields, the $z$-component of vectors, and the $rr$ and $zz$ components of 2-tensors must be even ($O(1)$) across the axis of symmetry. All other variables are odd ($O(r)$) across this axis. Finally, symmetric 3-tensors $M_{ij}$ must satisfy the condition

$$\frac{M_{\phi\phi}}{r^2 M_{rr}} = 1 + O(r^2) \tag{30}$$

near the axis, so we introduce three new dynamical variables $s, Y, \tilde{\tau}$ and perform a change of variables.

$$\begin{aligned} \lambda &= r e^{rs} \sqrt{h_{rr}} \\ L &= \frac{X_{rr}}{h_{rr}} + rY \\ \tau &= \frac{S_{rr}}{h_{rr}} + r\tilde{\tau} \end{aligned} \tag{31}$$

Here $s$ is known as the seed function, and all three scalar fields are odd across the axis. Evolving these variables instead of $\lambda, L$, and $\tau$ ensures the regularity condition is satisfied at all times. This also allows us to eliminate singular terms such as $\lambda^{-1} D_A \lambda$ in the evolution equations via L'Hôpital's rule.

The axisymmetric simulations referenced here were further restricted to a single quadrant for optimal efficiency. This requires implementing parity boundary conditions across the $z = 0$ axis, which ends up being very similar to $r = 0$ case. The only difference is the $r$-component of vectors is even across the $z = 0$ axis, whereas the $z$-component is odd.

### E. Initial Data

We assume a moment of time symmetry at the initial time step. This sets $X_{AB} = L = 0$. Constraints should be satisfied exactly initially, which implies $\Theta = Z_A = 0$. The gauge conditions are set to $\alpha = 1$, $\beta_A = 0$. We assume an initial line element of the form $dl^2 = \psi^4 \left(e^{-2rs}(dr^2 + dz^2) + r^2 d\phi^2\right)$ [16].

$$\begin{aligned} \sqrt{h} &= \psi^4 e^{-2rs} \\ \lambda &= r\psi^2 \end{aligned} \tag{32}$$

Under these conditions the Hamiltonian constraint becomes

$$\begin{aligned} \partial_r^2 \psi + \frac{\partial_r \psi}{r} + \partial_z^2 \psi - \frac{\psi}{4}(r\partial_r^2 s + 2\partial_r s + r\partial_z^2 s) \\ + \frac{\kappa\rho}{4} e^{-2rs}\psi^5 = 0. \end{aligned} \tag{33}$$

which can be solved for the conformal factor given some initial seed function and energy-momentum term. This is then used to set initial values for the components of the metric.

To simulate Brill waves we choose a common seed function first proposed by Holz, Miller, Wakano and Wheeler [17] and studied in Refs. [7, 16, 18], which takes the form

$$s(r, z, \phi) = -A_s r \exp\left[-\frac{r^2}{\sigma_r^2} - \frac{z^2}{\sigma_z^2}\right] \tag{34}$$

for some amplitude $A_s$.

### F. Energy-Momentum Sources

In this work we restrict ourselves to real-valued scalar fields as our model of energy-momentum. These obey the Klein-Gordon equation

$$\nabla^a \nabla_a \varphi = m^2 \varphi. \tag{35}$$

We can reduce this to first order in time by defining $\pi \equiv \mathcal{L}_{\mathbf{n}} \varphi$, and then perform the ADM and Geroch decompositions to derive axisymmetric evolution equations.

$$\begin{aligned} \mathcal{L}_{\mathbf{n}} \pi =& D_A D^A \varphi + \pi(X + L) \\ &+ \frac{D_A \varphi D^A \alpha}{\alpha} + \frac{D_A \varphi D^A \lambda}{\lambda} - m^2 \varphi \\ \mathcal{L}_{\mathbf{n}} \varphi =& \pi \end{aligned} \tag{36}$$

Similarly, using the energy-momentum tensor for a scalar field

$$T_{ab} = \nabla_a \varphi \nabla_b \varphi - \frac{1}{2} g_{ab} (\nabla_c \varphi \nabla^c \varphi + m^2 \phi^2) \tag{37}$$

we can compute the axisymmetric energy-momentum terms needed for Eqs. (29).

$$\begin{aligned} \rho =& \frac{1}{2}(\pi^2 + D_A \varphi D^A \varphi + m^2 \varphi^2) \\ \mathcal{J}_A =& -\pi D_A \varphi \\ \mathcal{S}_{AB} =& D_A \varphi D_B \varphi \\ &+ \frac{1}{2} h_{AB}(\pi^2 - D_C \varphi D^C \varphi - m^2 \varphi^2) \\ \tau =& \frac{1}{2}(\pi^2 - D_A \varphi D^A \varphi - m^2 \varphi^2) \end{aligned} \tag{38}$$

### G. Gauge Conditions

We tested the efficiency of several different gauge conditions before running full critical searches. For the lapse function these conditions included log+1 slicing [19]

$$\partial_t \alpha = -2\alpha(X + L - 2\Theta) + \beta^A \partial_A \alpha, \tag{39}$$

and harmonic slicing [20]

$$\partial_t \alpha = -\alpha^2(X + L - 2\Theta) + \beta^A \partial_A \alpha. \tag{40}$$

For shift conditions, we either assume zero shift or pair our harmonic slicing with a harmonic shift condition.

$$\begin{aligned} \partial_t \beta^A =& -\alpha^2 \left[\partial^A \ln(\lambda\sqrt{h}) + \partial_B h^{AB} - 2Z^A + \partial^A \ln\alpha\right] \\ &+ \beta^B \partial_B \beta^A + \alpha^2 G^A \end{aligned} \tag{41}$$

$G^A$ is a regularity term which we define as follows.

$$G^r = r^{-1} h^{rr}, \;\; G^z = 0 \tag{42}$$

The harmonic shift condition in axisymmetry is derived in Ref. [15] and all gauge conditions were empirically found to have slightly varying efficiencies when it comes to resolving critical phenomena. For instance, harmonic slicing resolves dispersing simulations faster, but takes about twice as long as log+1 slicing to resolve collapse simulations. Accordingly, for all critical searches in this work we use log+1 slicing.

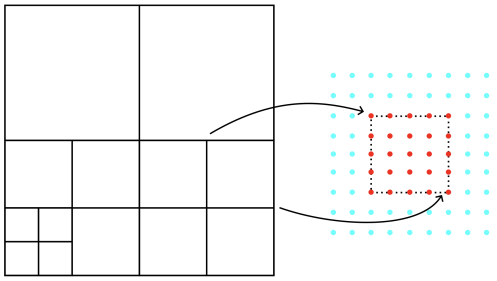


FIG. 1. Example adaptive mesh ($d = 4$), along with the physical nodes (red) and ghost nodes (teal) associated with a single cell.

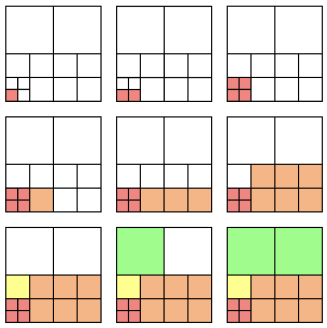


FIG. 2. Diagram of greedy meshing algorithm, proceeding left-to-right, top-to-bottom. Different colors denote different blocks.

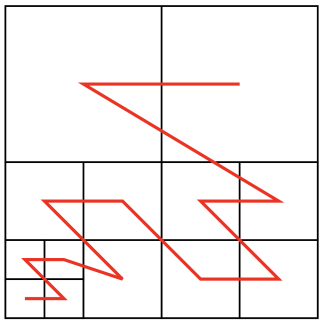


FIG. 3. Z-order curve associated with an example adaptive mesh.

## III. NUMERICAL METHODS

To solve this coupled system of PDEs we implement an adaptive, finite difference code known as `aeon` [1]. In order to handle the varying spatio-temporal scales of structure that arise in critical phenomena simulations, as well as ensure convergence in regions with high gradients, we utilize an adaptive, non-uniform mesh that is updated every few timesteps. The code represents the axis-aligned domain as a quadtree of cells with increased depth in regions requiring higher refinement. Each cell contains a uniform grid of $(d+1)^2$ nodes on which we store function values, where $d \geq 1$ is a fixed parameter controlling the width of each cell, and is further padded by $d/2$ ghost nodes along each face. The values of these ghost nodes are either interpolated or copied from neighboring cells or set via physical boundary conditions along the edges of the domain. Thus we can use standard, uniform, centered finite difference stencils to compute spatial derivatives on each cell in parallel. To allow interpolation between neighboring cells with precomputed stencils we require that neighboring cells (across edges and corners) are always within one level of refinement of one another, which enables cleaner parallelization and interpolation between neighboring cells.

In order to achieve fourth-order accuracy in spatial differentiation we need $d \geq 4$. However, Kreiss-Olgier dissipation [21] (required for our explicit temporal integrator) requires centered stencils with 7 support points total, so $d$ is set to 6 for all simulations in this work.

Including too many redundant ghost points for each cell is wasteful, as they often end up outnumbering the physical nodes. Furthermore, inter-cell synchronization is already one of the most time-consuming parts of each timestep. To avoid this inefficiency, we group cells of the same level of refinement into uniform rectangular blocks using a clustering algorithm inspired by greedy meshing in graphics programming [22]. The code loops over all active cells in the quadtree. If a cell is not already associated with a block, then it initializes a new block. For each axis, it attempts to expand the block to contain neighboring cells along that axis, succeeding if those neighboring cells have the same level of refinement and are not already associated with a block (see Fig. 2). The final list of blocks subdivides the domain into uniform rectangular regions of nodes, on which we can compute spatial derivatives.

Similar to `bamps` [23] the quadtree is stored as a list of leaves, which are ordered by a Z-order curve (see Fig. 3). The underlying tree structure is stored using each cell's Morton code [24], which allows us to perform refinement/coarsening as local operations within this list, and provides a natural way to associate cells to different threads/processors while still largely preserving spatial coherence.

In order to adapt the domain to the physical problem at hand we need a criterion for determining whether a given cell should be coarsened or refined. Based on Dendro GR [25], our codebase uses a version of wavelet adaptive mesh refinement where each cell is expanded in terms of interpolating wavelets with one less level of refinement and detail coefficients are computed on intermediate nodes (see Fig. 4). If the largest detail coefficient is larger than $10^{-6}$ that cell is marked for refinement, and if it is smaller than $10^{-10}$ that cell is marked for coarsening. In either case we perform a post-processing step to make sure the 2:1 refinement condition is still satisfied

[1] Available online at `https://github.com/sylviamesicek/aeon`

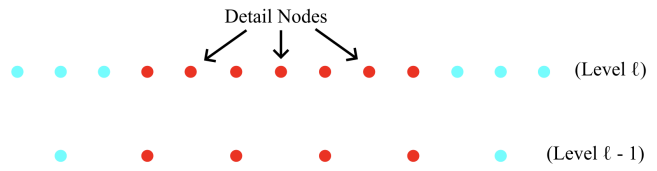



FIG. 4. Fourth-order interpolating wavelets are applied to level $\ell-1$ nodes on the cell. The difference between the values of this wavelet representation and actual function values at detail nodes (nodes on level $\ell$ that are not on level $\ell-1$) are called *detail coefficients*.

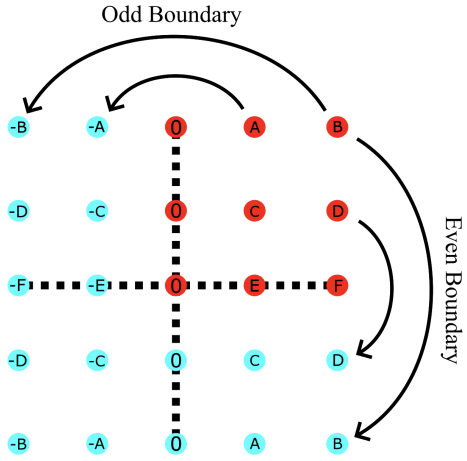



FIG. 5. Example of enforcing a even-parity condition across the $r$-axis and an odd-parity condition across the $z$-axis.

after regenerating the mesh.

Strong boundary conditions (for instance parity, Dirichlet or Neumann type conditions) are enforced by fixing the values of a function on or near a boundary prior to finite differencing. In cases where the value of a function is fully specified outside of the numerical domain, we can continue to use centered finite difference stencils near the boundary, otherwise we switch to the appropriate one-sided stencil. On the $r$-axis and $z$-axis boundaries, fields can either have odd or even parity. In the case of odd-parity, we fix the value of the field to 0 along the boundary and set ghost nodes to the negated values of corresponding physical nodes, while in the case of even-parity, we simply copy the values of physical nodes to their corresponding ghost nodes (Fig. 5). This method lets wave fronts cleanly reflect/move across the inner boundary, and allows us to simulate a single quadrant of the numerical domain to save on computational time. Finally, the regularity of Eqs. (29) assumes certain fields either vanish or have zero derivative across the $z$-axis. When these conditions are not strongly enforced, the simulation quickly crashes due to instabilities/exponentially growing modes near this coordinate singularity.

On the outer boundary, we use a Sommerfeld radiative boundary condition: a weak boundary condition that enables outgoing spherical waves to pass through the boundary with minimal reflection [26]. Following the derivation in Ref. [27] this condition implements the ansatz

$$f(t,R) = f_0 + \frac{w(R-ct)}{R} + \frac{C}{r^2} \tag{43}$$

by setting the time derivative on the boundary

$$\partial_t f = -c\left(\partial_R f + \frac{f-f_0}{R}\right) + \frac{k}{R^3}, \tag{44}$$

where $c$ is the wave speed (set to 1 in our units) and $k$ captures higher order corrections between our hyperbolic PDE and the spherical advection equation we are implementing on the boundary. We can compute $k$ by evaluating the difference between these two PDEs at the interior node closest to the boundary node.

$$k = R_{\rm int}^3\left(\partial_t f + c\left(\partial_R f + \frac{f-f_0}{R}\right)\right)_{\rm int} \tag{45}$$

After computing spatial derivatives using the methods above, we integrate the PDEs forward in time using a standard, fourth-order Runge-Kutta method. Using this explicit integrator naively yields numerical instabilities, as dynamical coefficients in Eq. (29) result in non-physical, growing modes on the scale of the grid spacing. To dampen these modes, we use sixth-order Kreiss-Olgier dissipation [21]. Note sixth-order dissipation is required to not spoil the fourth-order convergence of the scheme [28].

After performing several time steps (by default 20) wavefronts may have traveled across existing blocks and into less refined regions of domain. As such we must run our regridding algorithm, and copy field data from the old mesh to the new mesh via injection and interpolation where appropriate. As a validation test, this codebase was first applied to spherically symmetric critical phenomena as the finite difference comparison code used in Ref. [29].

### A. Solving Elliptic PDEs

In order to solve the elliptic PDEs which must be satisfied on the initial time-slice in our formalism, we turn to hyperbolic relaxation, specifically implementing the method implemented in the NRPyElliptic codebase [27]. This method transforms an elliptic equation of the form

$$\mathcal{L}_E\vec{u} - \vec{p} = 0 \tag{46}$$

by introducing a fictitious time coordinate $t$ and integrating

$$\partial_t^2\vec{u} + \eta\partial_t\vec{u} = c^2\left(\mathcal{L}_E\vec{u} - \vec{p}\right) \tag{47}$$

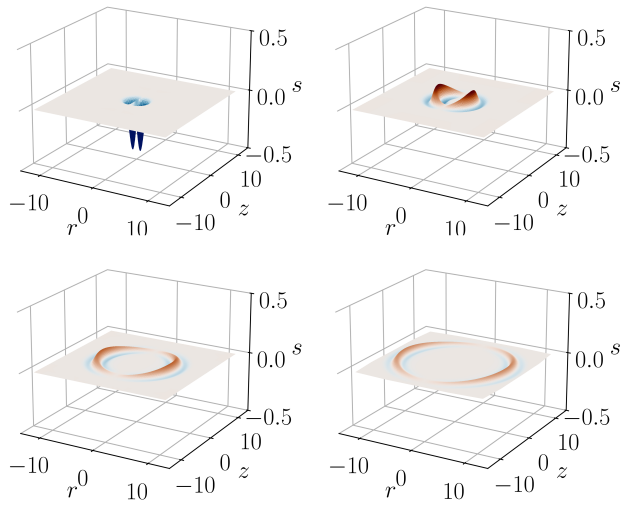



FIG. 6. Simulation of Brill waves with $A_s = 1$ and harmonic gauge conditions. The simulation proceeds left to right, top to bottom with slices from $\tau = 0.0, 3.3, 6.6, 10.0$ (where $\tau$ is proper time measured by an observer at the origin). The seed function is plotted on the $z$-axis.

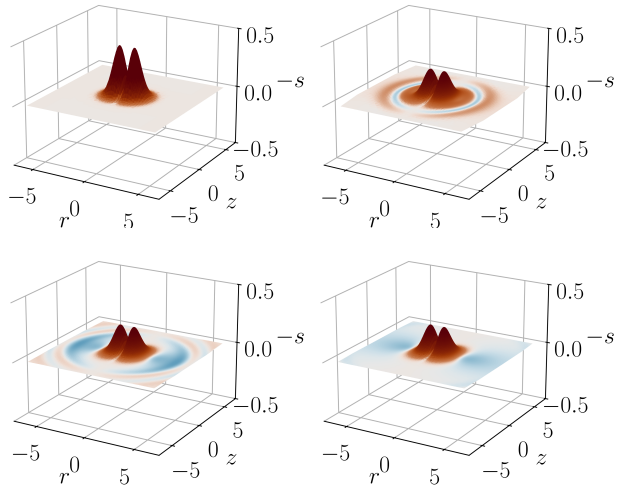



FIG. 7. Simulation of Brill waves with $A_s = 1$, harmonic slicing, and zero shift. The simulation proceeds left to right, top to bottom with slices from $\tau = 0.0, 3.3, 6.6, 10.0$. For clarity the negative seed function is plotted on the vertical axis to demonstrate gauge freezing of the solution.

until a steady state solution is found. Here $\eta$ is the exponential dampening parameter (set to 0.4 for our initial data solver), and $c$ is the wave speed (set to 1 in natural units). Reducing Eq. 47 to first order in time creates a hyperbolic system that can be integrated via the method of lines.

$$\begin{aligned} \partial_t \vec{u} &= \vec{v} - \eta \vec{u} \\ \partial_t \vec{v} &= c^2 \left( \mathcal{L}_E \vec{u} - \vec{p} \right) \end{aligned} \tag{48}$$

Relaxation terminates when $||\mathcal{L}_E \vec{u} - \vec{p}||_{L^2} < 10^{-6}$. We flag the solution for regridding, and repeat relaxation on a more refined mesh until no cells trigger the wavelet refinement condition.

This formulation bypasses the complexities of a full elliptic solver by allowing us to apply existing infrastructure in the codebase designed for integrating hyperbolic equations to solve for initial data. It also avoids the computational pitfalls of a scheme like parabolic relaxation which suffers from a $\Delta t \sim (\Delta x)^2$ stability requirement [30].

## IV. RESULTS

All simulations were performed on a $48 \times 48$ domain representing the positive $r$, positive $z$ quadrant. We use a Gaussian profile for the initial massless scalar field

$$\varphi = A_\Phi \exp\left[ -\frac{r^2}{5.35^2} - \frac{z^2}{5.35^2} \right] \tag{49}$$

and superimpose a Brill wave by varying the seed amplitude $A_s$ in Eq. 34 (fixing $\sigma_r = \sigma_z = 1$). Thus setting

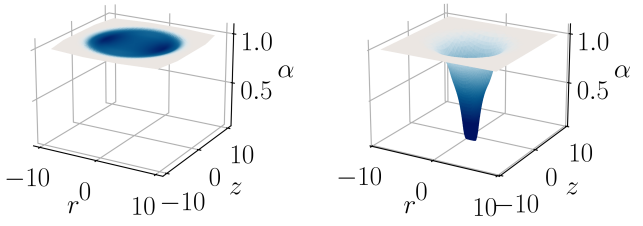



FIG. 8. Lapse function for a dispersing $A_s = 1$ (left) vs collapsing $A_s = 5$ (right) Brill wave. The dispersing wave is displayed at $\tau = 5.0$, while the collapsing wave is displayed at $\tau = 0.78$. Simulations were performed using log+1 slicing and zero shift for computational efficiency.

$A_\Phi = 0$ corresponds to pure vacuum collapse (i.e. $\rho = 0$) and fixing $A_s = 0$ corresponds to a standard spherically symmetric scalar field. To solve for the initial metric we compute $\rho$ by inserting Eq. 49 into Eq. 38, and use the resulting value in Eq. 33 to derive a cylindrical Helmholtz equation with an additional nonlinear $\psi^5$ term. We use log+1 slicing with zero shift (Eq. 39). By altering $A_\Phi$ and $A_s$ we can explore the parameter space of this coupled system.

Figure 6 demonstrates the evolution of a dispersing Brill wave with a harmonic gauge condition. The seed function $s$ can be roughly interpreted as the transverse degree of freedom of the gravitational wave. In this case, the wavefront expands outwards and dissipates, approaching a static Minkowski spacetime as $t \to \infty$. Per-

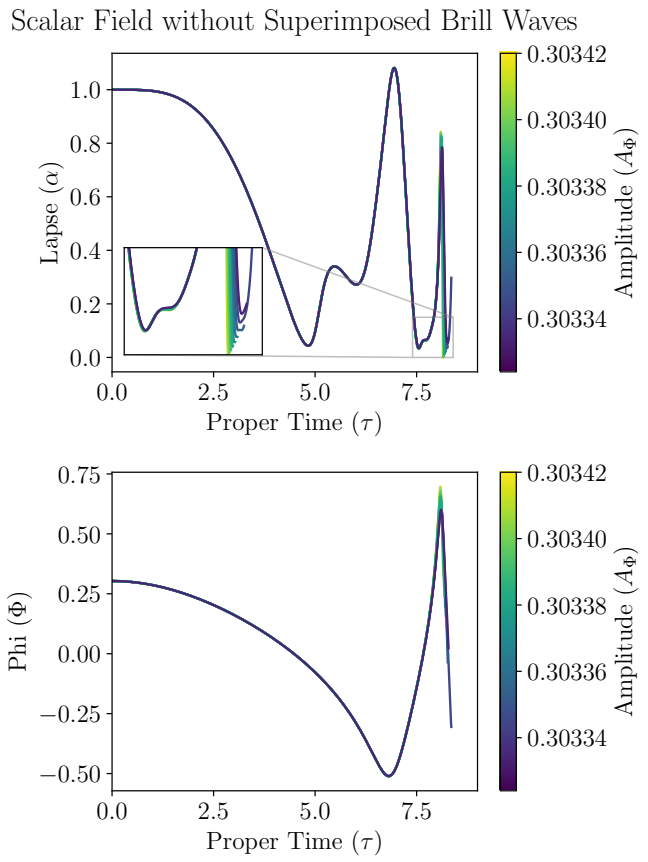



FIG. 9. Finest critical search iteration for a massless scalar field $A_s = 0$. Color represents the amplitude $A_\Phi$ used for the initial state of the system, and the x-axis is proper-time as measured by a central observer. The top plot shows lapse at the origin, whereas the bottom panel shows the value of the scalar field at the origin

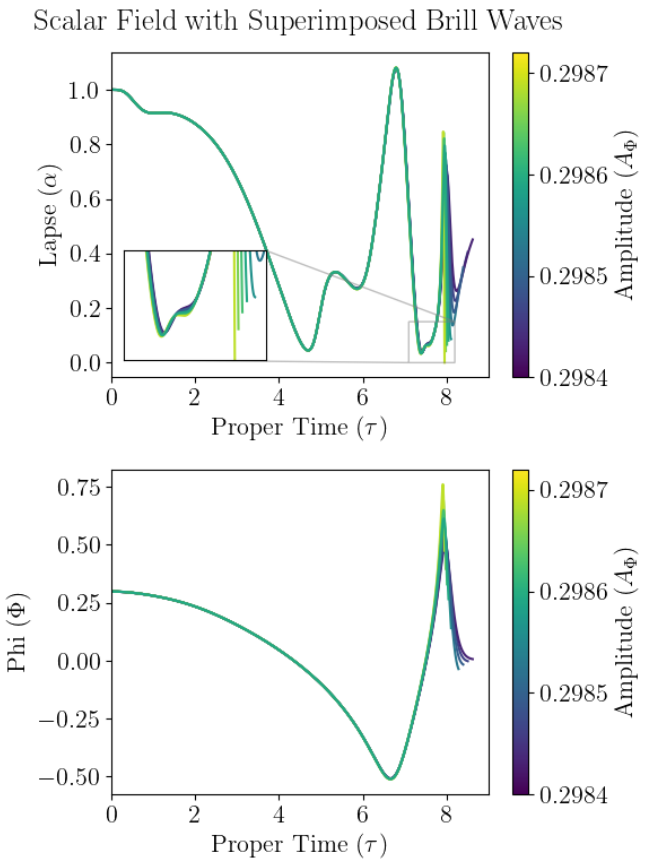



FIG. 10. Finest critical search iteration for a system consisting of a superimposed massless scalar field and Brill wave. The meaning of axes is the same as in Fig. 9.

forming the same simulation with vanishing shift yields a non-trivial static solution consistent with the results observed by Ref. [15] as demonstrated in Fig. 7. This solution is still Minkowski spacetime, albeit in nonstandard coordinates containing a "lump of gauge" near the origin. For gauge conditions with vanishing shift, the difference between dispersing and collapsing spacetimes can be seen in the lapse function (Fig. 8), which approaches either 1 (dispersion) or 0 (collapse) at the origin.

In Figs. 9 and 10 we demonstrate critical searches in $A_\Phi$ for $A_s = 0$ and $A_s = 1$ respectively. These searches proceed as follows. For a fixed $A_s$ and scalar field amplitudes in the range $A_\Phi \in [A_{\text{disperse}}, A_{\text{collapse}}]$ we can perform a decenary search to find the critical point. An evolution is marked as dispersing if it reaches a fixed post-critical proper-time $\tau = 9.5$, and is marked as collapsing if the curvature diverges, the lapse drops below $10^{-8}$ at the origin, or the evolution runs for a certain number of steps (in practice this heuristic must be adjusted as we get closer to the critical point to prevent dispersing evolutions from being erroneously classified as collapsing). The scalar field solution was tuned to within $\approx 10^{-5}$ of the critical value whereas the superimposed solution was tuned to within $\approx 10^{-4}$ of the critical value. This close to criticality, each evolution takes approximately 150 core-hours to distinguish between dispersing and collapsing systems, meaning each iteration of the decenary search takes around 1500 core-hours.

The addition of the Brill wave introduces energy-momentum into the system which appears to interact constructively with the scalar field, thus reducing the critical point by $\approx 0.005$ and reducing the critical collapse time. With this degree of criticality we observe a second echo (and the beginning of a third echo) in $\alpha$ as well as the beginning of a second echo in $\phi$. The contribution of the Brill wave to the overall dynamics of evolution (visible around $0 < \tau < 1$ in Fig. 10) are quickly damped out by the scalar field, resulting in near-identical evolution when close to the critical collapse time.

## V. CONCLUSION & FUTURE WORK

We have derived a hyperbolic mathematical formulation for competing gravitational waves and scalar fields

in axisymmetric GR, and presented a publicly available numerical codebase, designed for robust simulations of a number of physical systems. We were able to evolve both dispersing and collapsing Brill waves, and observed behavior consistent with critical phenomena for superimposed scalar field and Brill wave collapse. Supporting the predictions of various fully symmetric models [8, 9] we find evidence that gravitational wave perturbations are dampened out by a near-critical scalar field.

This codebase serves as an important stepping stone for our higher-dimensional competing collapse studies, unlocking a number of axes of the parameter space of competing energy-momentum sources (including prolate scalar fields [31] and alternate gravitational wave seed functions). Weak Brill waves appear to work constructively with scalar field collapse rather than competitively. However, we postulate that, given computational time to tune the Brill wave perturbations to criticality, we would observe a similar intersection of critical surfaces as seen in Ref. [10] wherein gravitational wave dynamics would dominate over scalar-field dynamics in certain regions of parameter space. In particular, whether the transition between matter-dominated and gravitational-wave-dominated collapse is governed by a distinct boundary solution or by a more complicated, family-dependent dynamics remains an open question for future study.


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